

Xinran Yu

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Education

University of Illinois Urbana-Champaign

May 2026

Ph.D. in Mathematics, Advisor: Prof. Pierre Albin

M.S. in Mathematics, May 2019

University of Liverpool & Xi'an Jiaotong-Liverpool University

June 2017

B.Sc. in Mathematics with Honors & B.Sc. in Applied Mathematics (dual degree, 2+2 program)

Thesis: *Analytic Continuation and Riemann Surfaces*

Advisor: Dr. Jon Woolf

Research Interests

Geometric Analysis, Conformal Geometry, Mathematical Relativity, Witten Complexes on Singular Spaces.

Preprints and Publications

1. Spectral asymmetry, supersymmetry and the equivariant Riemann-Roch defect
with Gayana Jayasinghe and Alex R. Taylor, [arXiv:2603.26057](#) 2026
2. Conformally Compact Metrics and the Lovelock Tensors
[arXiv:2505.24188](#), submitted 2025
3. Witten Instanton Complex and Morse-Bott Inequalities on Stratified Pseudomanifolds
with Gayana Jayasinghe and Hadrian Quan, [arXiv:2412.12003](#), submitted 2024

Talks

- Workshop 26w5550, Banff International Research Station (Invited) Apr 2026
“Conformally Compact Metrics and the Lovelock Tensors”
- AMS Fall Central Sectional Meeting, Saint Louis University (Invited) Oct 2025
“Asymptotic Behavior of Conformally Compact Lovelock Metrics”
- Geometric Analysis Seminar, UIUC (Invited) Sep 2025
“Fefferman-Graham expansion of Conformally Compact Lovelock Metrics”
- Great Lakes Mathematical Physics Meeting (GLaMP), University of Kentucky Jun 2025
“Asymptotic Behavior of Conformally Compact Lovelock Metrics”
- Graduate Student Geometry and Topology Seminar, UIUC Oct 2024
“Dirichlet-to-Neumann Map for Conformally Compact Einstein Metrics”
- AWM Graduate Student Colloquium, UIUC Apr 2024
“Conformal Geometry in Lovelock Gravity Theories”

Honors & Awards

- Boya Postdoctoral Fellowship 2026
- Susan C. Morisato IML Scholarship, UIUC Summer 2025
- Ruth V. Shaff & Genevieve I. Andrews Fellowship, UIUC Spring 2024
- Teachers Ranked as Excellent by their Students, UIUC 2021
- Wills Prize in Mathematics, UoL 2017
- IMA Prize, UoL 2017
- University Academic Achievement Award, XJTLU 2014–2015

Teaching Experience

At UIUC

2019–2026

Instructor

- MATH 231: Calculus II (Spring 2025, Spring 2022)

Teaching Assistant

- MATH 241: Calculus III (Spring 2026, Fall 2021). Teachers Ranked as Excellent
- MATH 220: Calculus (Fall 2024)
- MATH 231: Calculus II (Spring 2021). Teachers Ranked as Excellent

Grader

Graduate Courses

- MATH 518–519: Differentiable Manifolds I–II (Fall 2023, Spring 2023)
- MATH 553: Introduction to Partial Differential Equations (Spring 2023)
- MATH 541: Functional Analysis (Spring 2023)
- MATH 512–514: Algebraic/Complex Algebraic Geometry (Fall 2022)
- MATH 525: Algebraic Topology I (Spring 2021)

Undergraduate Courses

- MATH 416: Abstract Linear Algebra (Fall 2025, Fall 2022)
- MATH 402, 403: Non-/Euclidean Geometry (Fall 2023)
- MATH 423: Differential Geometry (Fall 2023)
- MATH 286: Introduction to Differential Equations Plus (Fall 2020)
- MATH 489: Dynamics and Differential Equations (Fall 2019)
- MATH 417, 418: Introduction to Abstract Algebra I (Spring 2019), II (Fall 2019)

Mentorship

At UIUC

Illinois Mathematics Lab, Mentor

2024–2025

Projects conducted in collaboration with the Mathematics Library

- *Historical Mathematical Models* (Spring 2025) — Supervised 5 undergraduates.
- *Documenting Historical Mathematical Models* (Fall 2024) — Supervised 4 undergraduates.

Project received the Nancy D. Anderson Undergraduate Research Award.

Illinois Geometry Lab, Mentor

Fall 2019

- *Simulating Multi-Soliton Solutions to NLS and KdV* — Supervised 4 undergraduates.

Services & Enrichment

At UIUC

German Translation Project, Mathematics Library

2024–2025

International TA Panelist

Aug 2022

At UoL

Staff-Student Liaison Committee

2016–2017